



System Architecture

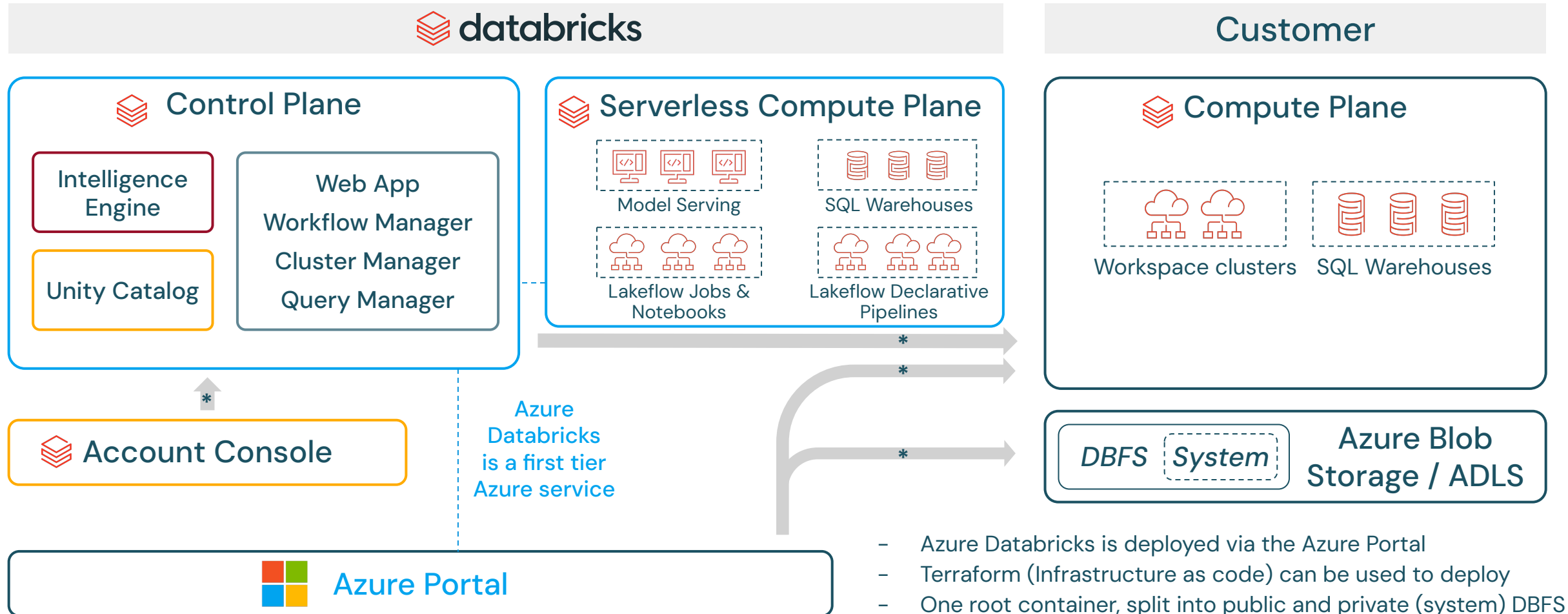
Azure Databricks

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Azure Databricks Deployment

With Unity Catalog



High Level Network Architecture

Network Access Options

User to Control Plane

- Public
 - the workspace URL is publicly accessible
 - access is restricted with IP Access Lists
- Private
 - only with front-end Private Link
 - must be allowed by the organization network (is a user's workstation able to reach this private endpoint?)
- Hybrid
 - front-end Private Link for most access
 - public with IP Access Lists for traffic from other cloud services that might not support Private Link (like Power BI Service)

VNet to Control Plane

- Default
 - the VNet is managed by Databricks
 - simple to deploy but no customizations
- VNet Injection
 - deploy Databricks in the customer's VNet
 - the VNet and subnets created in advance
 - cluster VMs receive Public IPs and the Control Plane connects to the VMs
- SCC (Secure Cluster Connectivity)
 - same as VNet Injection, but only Private IPs
 - traffic is now outbound (the cluster VMs connect to the Control Plane)
- Private Link
 - back-end Private Link (for the connection between VNet and Control plane)

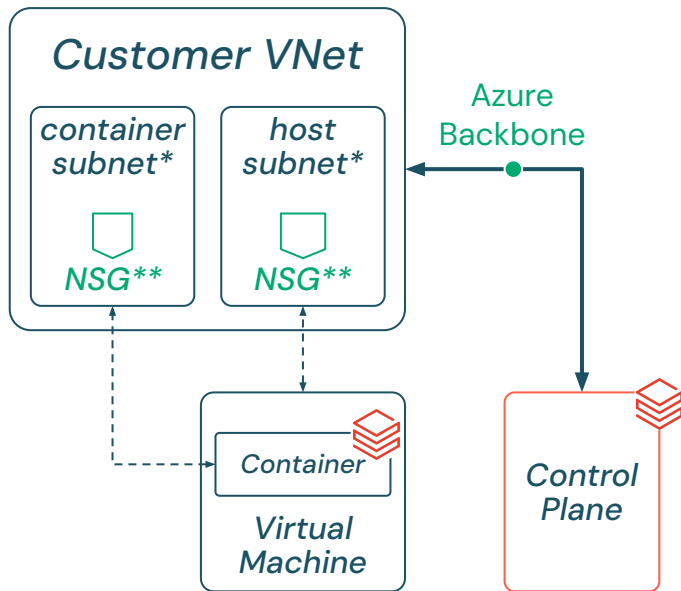
VNet to Storage

- Public Endpoints
 - the default -> storage accounts are accessible from the Internet
 - when used with SCC, the NAT Gateway can have a very high cost
- Service Endpoints
 - recommended with SCC as it bypasses the NAT Gateway and has no cost
 - however, additional configuration for each storage account is needed
- Private Endpoints
 - uses Private Link for data traffic between cluster VMs and the storage accounts
 - can reach very high costs

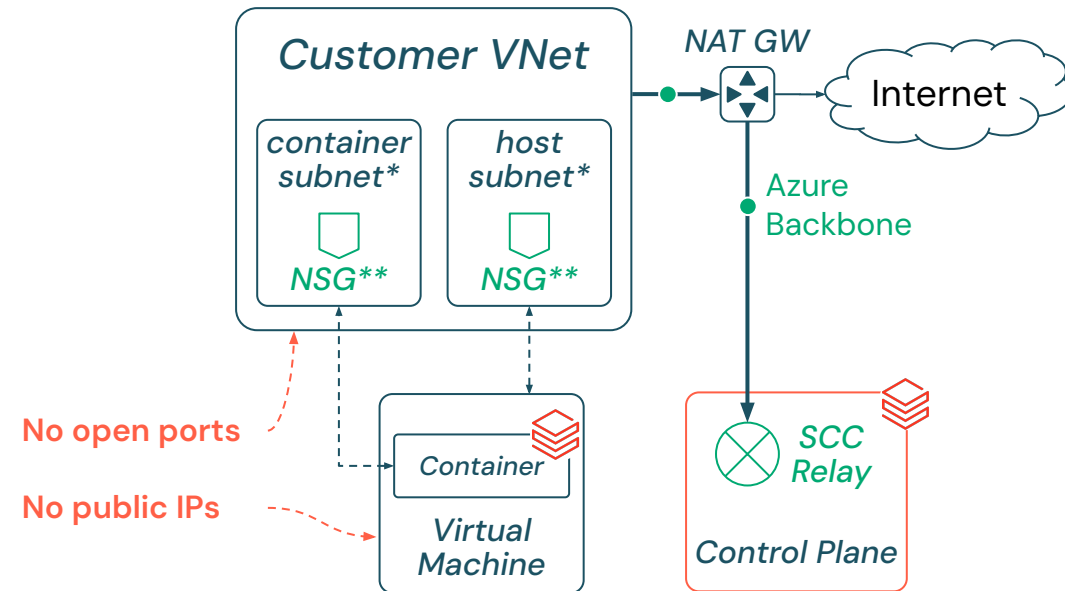


Azure Databricks Network Architecture

Bring Your Own Vnet (VNet Injection)



VNet injection + Secure Cluster Connectivity (SCC)



- The default setup uses Databricks managed VNet, however **VNet injection with SCC is recommended**
- **Vnet Injection** : Deploy Azure Databricks workspace in your own Virtual Network (VNet). Important features are full access to the NSGs and custom UDRs.
- **Secure Cluster Connectivity (SCC)**: No public IPs bound to VMs, and network traffic between the cluster and Control Plane goes outbound from the cluster to the Control Plane.
- A NAT Gateway is recommended so that the workspaces have a stable IPs when accessing service outside of Databricks

* "container subnet" is also referred to as "private subnet" and "host subnet" as "public subnet". Since the terms "private" and "public" are misleading, "container" and "host" are preferred

** managed by the customer

IP address ranges of Azure Databricks

Necessary IP ranges (subnets) for VNet injection

- VNet /16 - /24 (e.g. /20)
- (H) Host ("public-subnet") up to /26 (e.g. /21)
- (C) Container ("private-subnet") up to /26 (e.g. /21)
- Public IP (for non-NPIP) Azure IP range

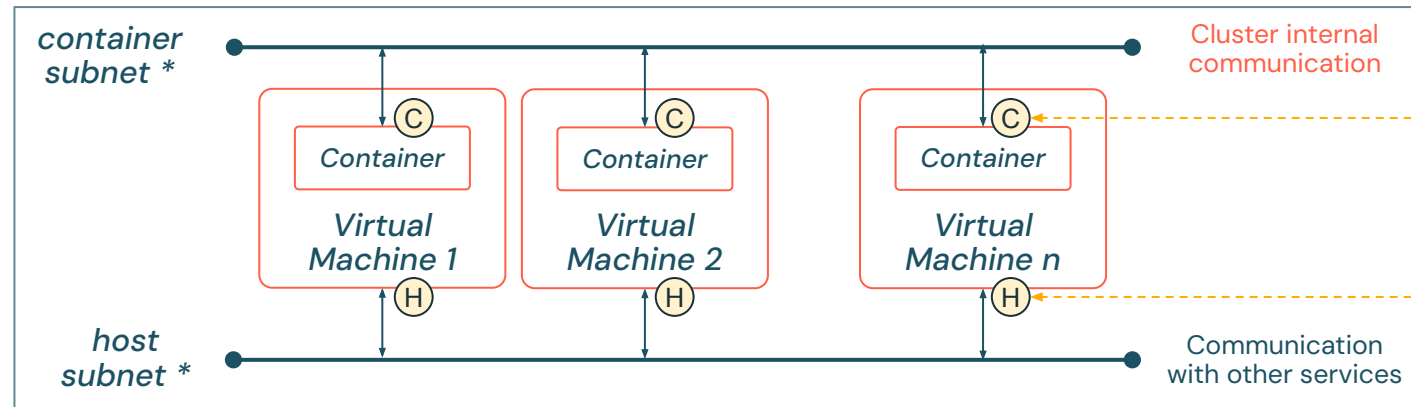
Notes:

- Private CIDR range size = Public CIDR range size
- Azure reserves 5 addresses for internal purposes, e.g. in a /20 VNet with /21 subnets the largest possible cluster would have one driver and maximal 2043 workers
- Even for "No Public IP" workspaces the host network is called "public-subnet" in the Azure Portal, however, it is a private IP range
- For network infrastructure nodes e.g. for Private Link separate subnets are needed. Consider to use smaller subnets than half of the VPC side (e.g. cidr - 2)

Other VNets



Customer managed Databricks Workspace VNet

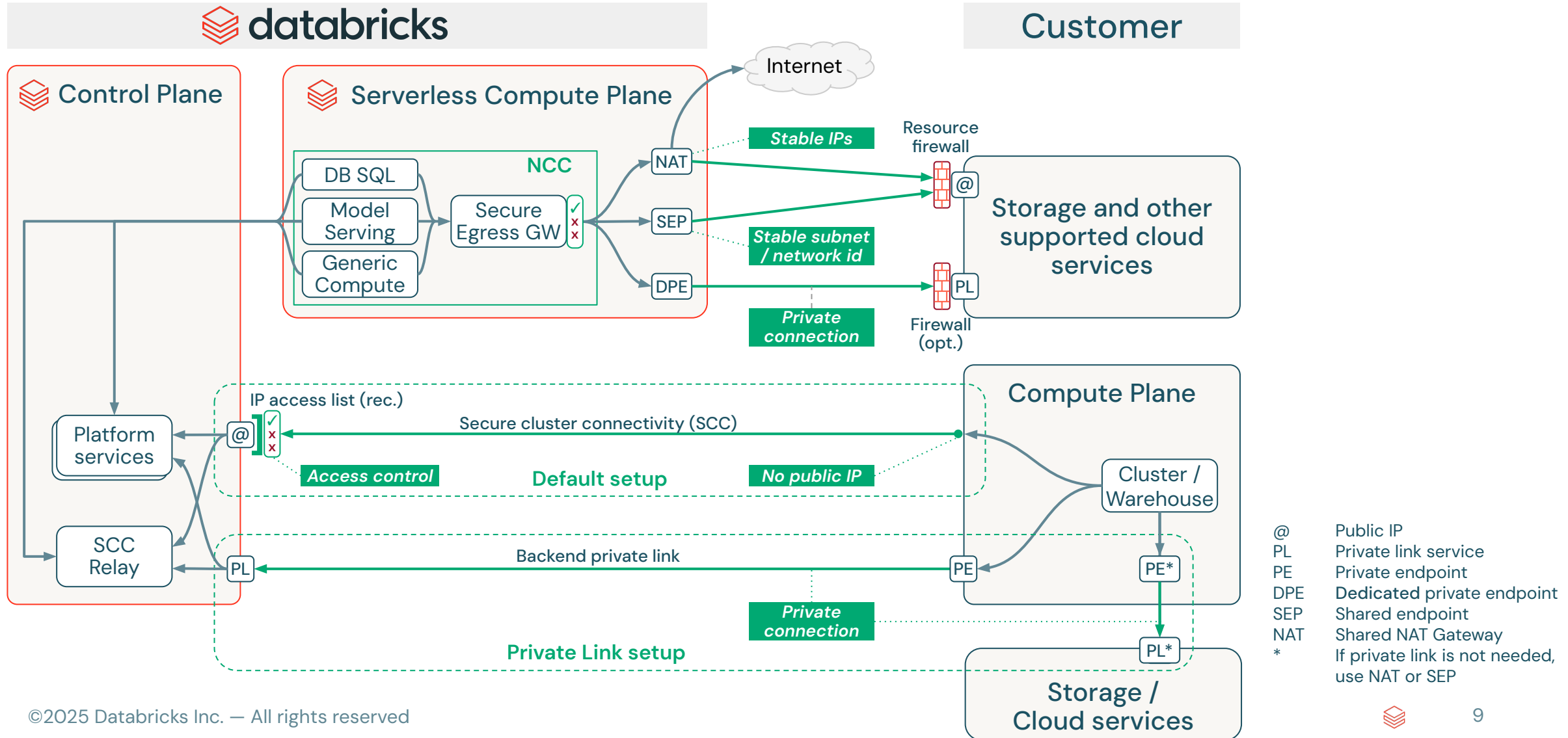


VM and container based Databricks cluster on Azure
(VMs and containers are managed by Databricks Control plane)

* "container subnet" is also referred to as "private subnet" and "host subnet" as "public subnet". Since the terms "private" and "public" are misleading, "container" and "host" are preferred

** Several options are supported, e.g Vnet Peering or User Defined Routes (UDR)

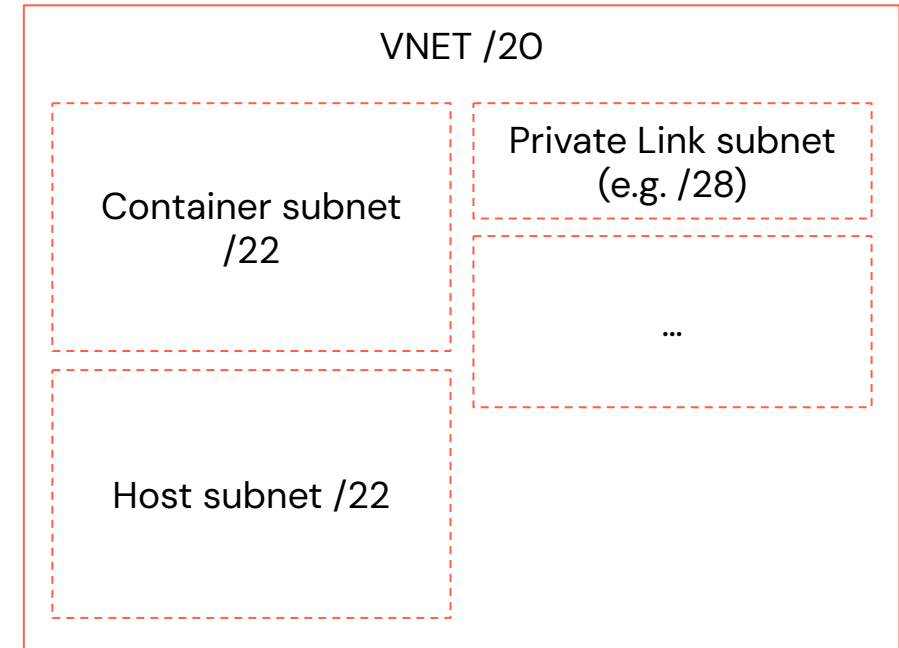
Databricks network controls



Recommended VNet setup

Being prepared for network security setups like Private Link

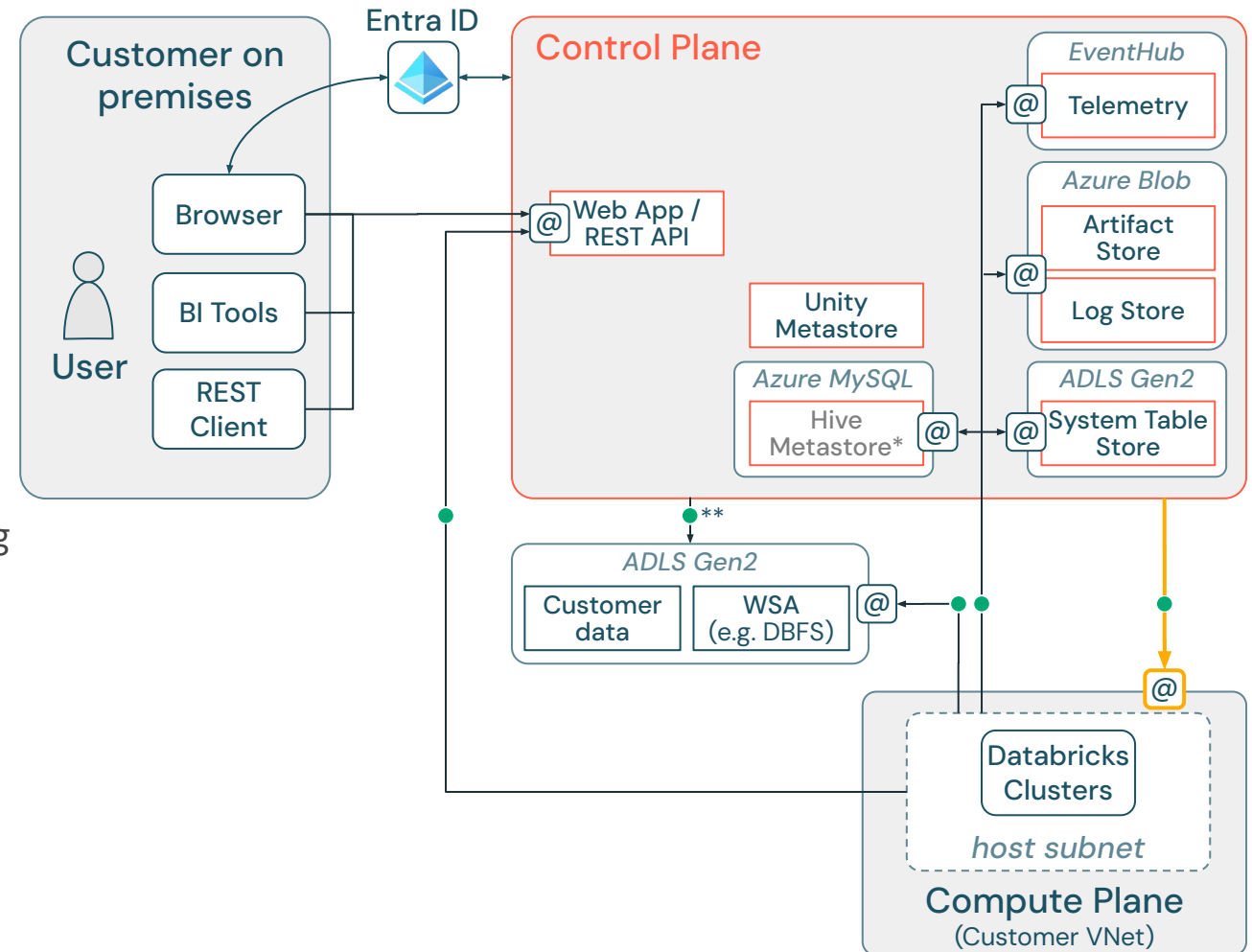
- It is possible to configure a network setup for Databricks with subnet size of Vnet CIDR - 1, e.g. VNet = /20, subnets = /21. This maximizes the possible cluster size, however there is no space in the VNet left for configuring more secure setups
- It is recommended to use VNet CIDR - 2 for the subnets, e.g. VNet = /20, subnets = /22
- Given this configuration, other subnets can be created in the VNet, e.g. to hold private endpoints for a private link configuration



Communication in a Databricks Workspace

- Compute Plane and Control Plane always communicate over Azure Backbone
- Control Plane to DBFS is going over Azure Backbone since both are first party Azure services
- Nevertheless, communication between Control and Compute Plane uses Public IPs on both planes
- Users access Databricks over the Public Internet

Note: Do NOT use DBFS as a storage layer (e.g. no access control) but rather use storage accounts with Unity Catalog access control for all customer tables, volumes, ...



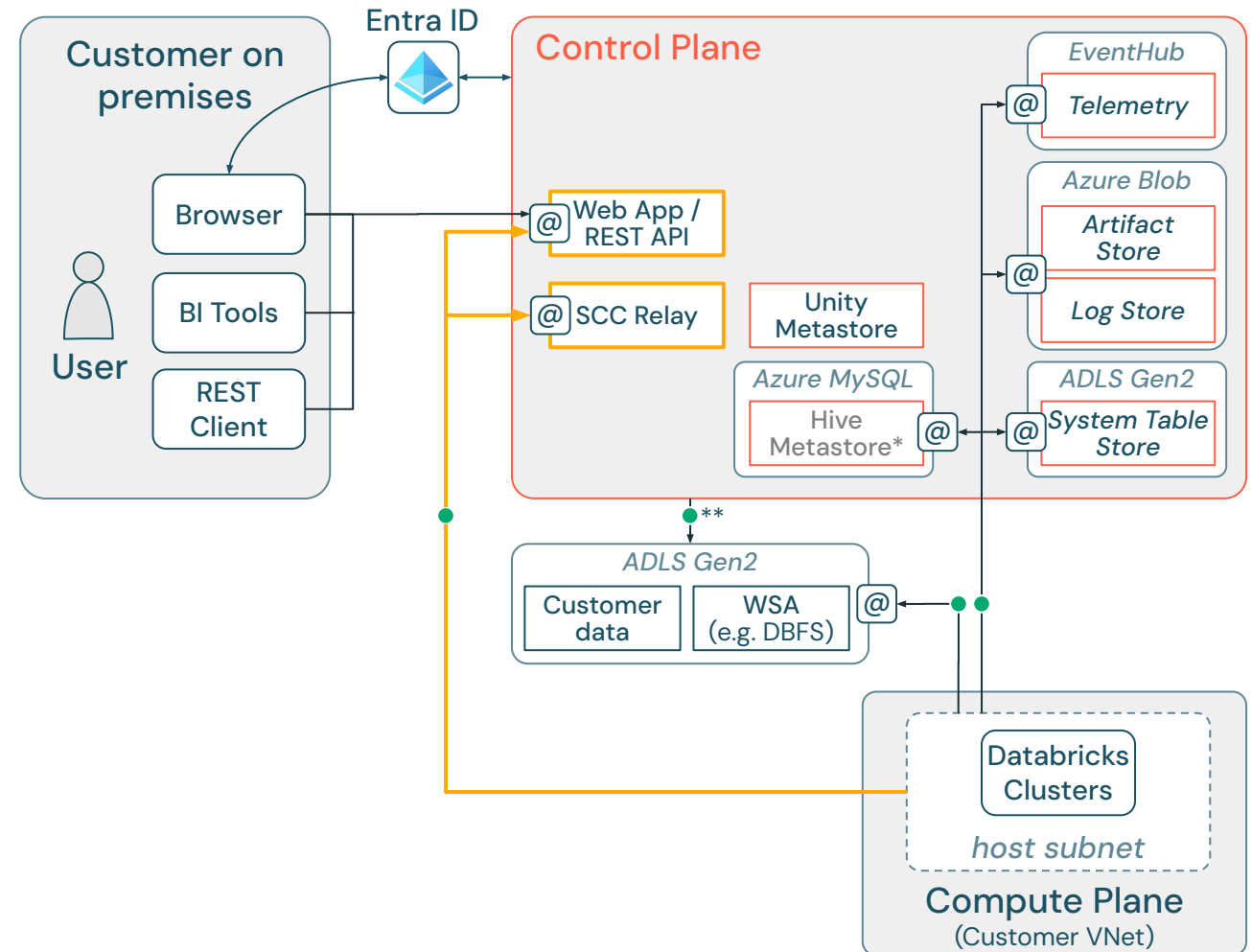
WSA = Workspace Storage Account

● = via Azure Backbone

@ = Public IP, TLS encrypted

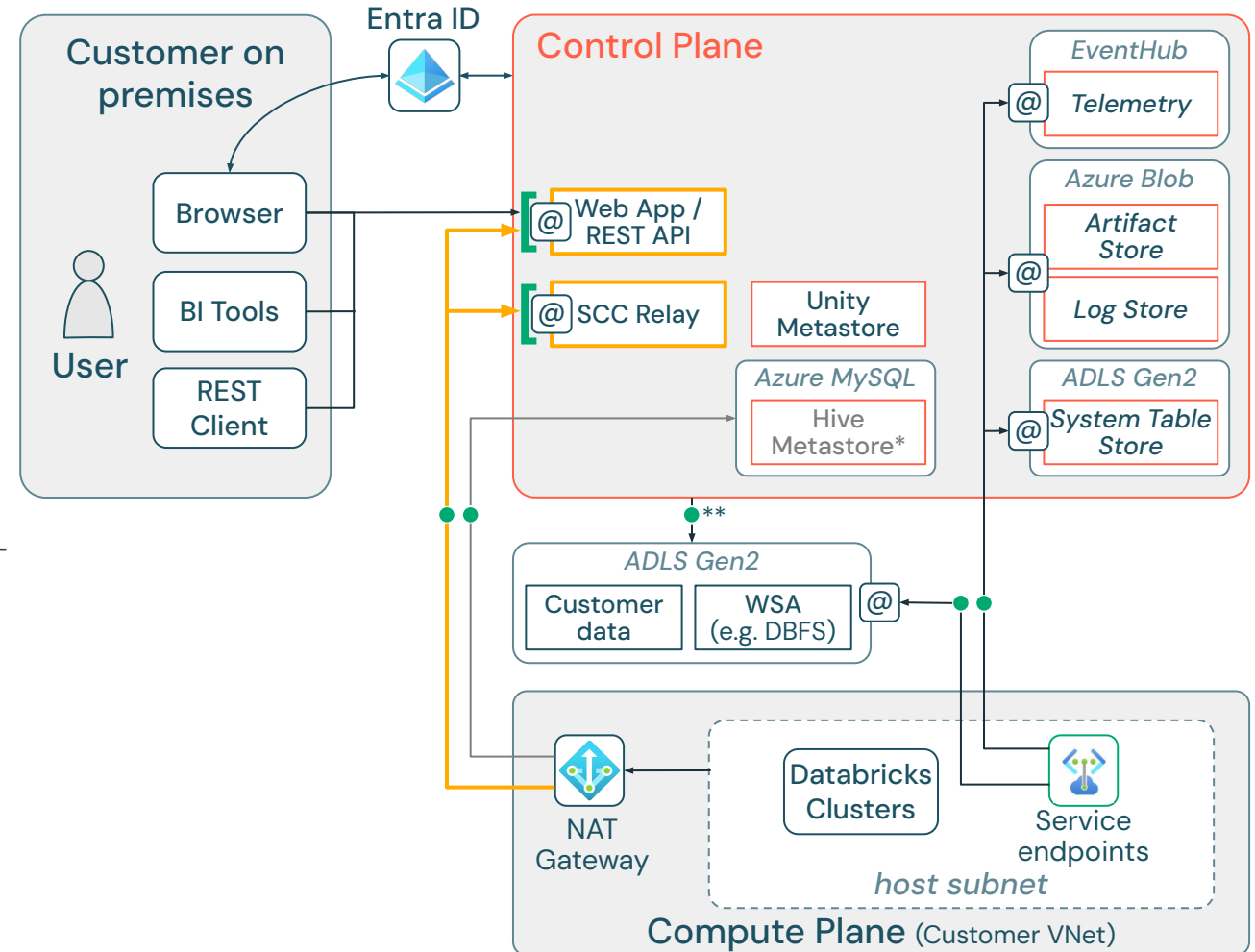
Communication in an SCC workspace

- It is recommended to deploy the workspace in customer's VNet (VNet injection)
- Secure Cluster Connectivity reverses the call direction via the SCC Relay and removes Public IPs from the Compute Plane.
- Calls from Compute Plane to the Control Plane still use Public IPs (use Private Link to remove the public IPs from the Control Plane, see next slides)



Communication in an SCC workspace with IP ACL

- The workspace needs to be deployed in customer's VNet (VNet injection)
- Secure Cluster Connectivity reverses the call direction via the SCC Relay and removes Public IPs from the Compute Plane.
- Calls from Compute Plane to the Control Plane still use Public IPs
- IP Access List defined in Control Plane restricts access to workspace from only allowed public IP CIDRs
- Create NAT Gateway in Customer VNET and allow NAT Gateway Public IP in IP Access List for compute plane to control plane access
- It is recommended to create service endpoint from Customer VNet to Microsoft.Storage to reduce NAT Gateway usage and reduce cost (and optional to Event Hub)



WSA = Workspace Storage Account

● = via Azure Backbone

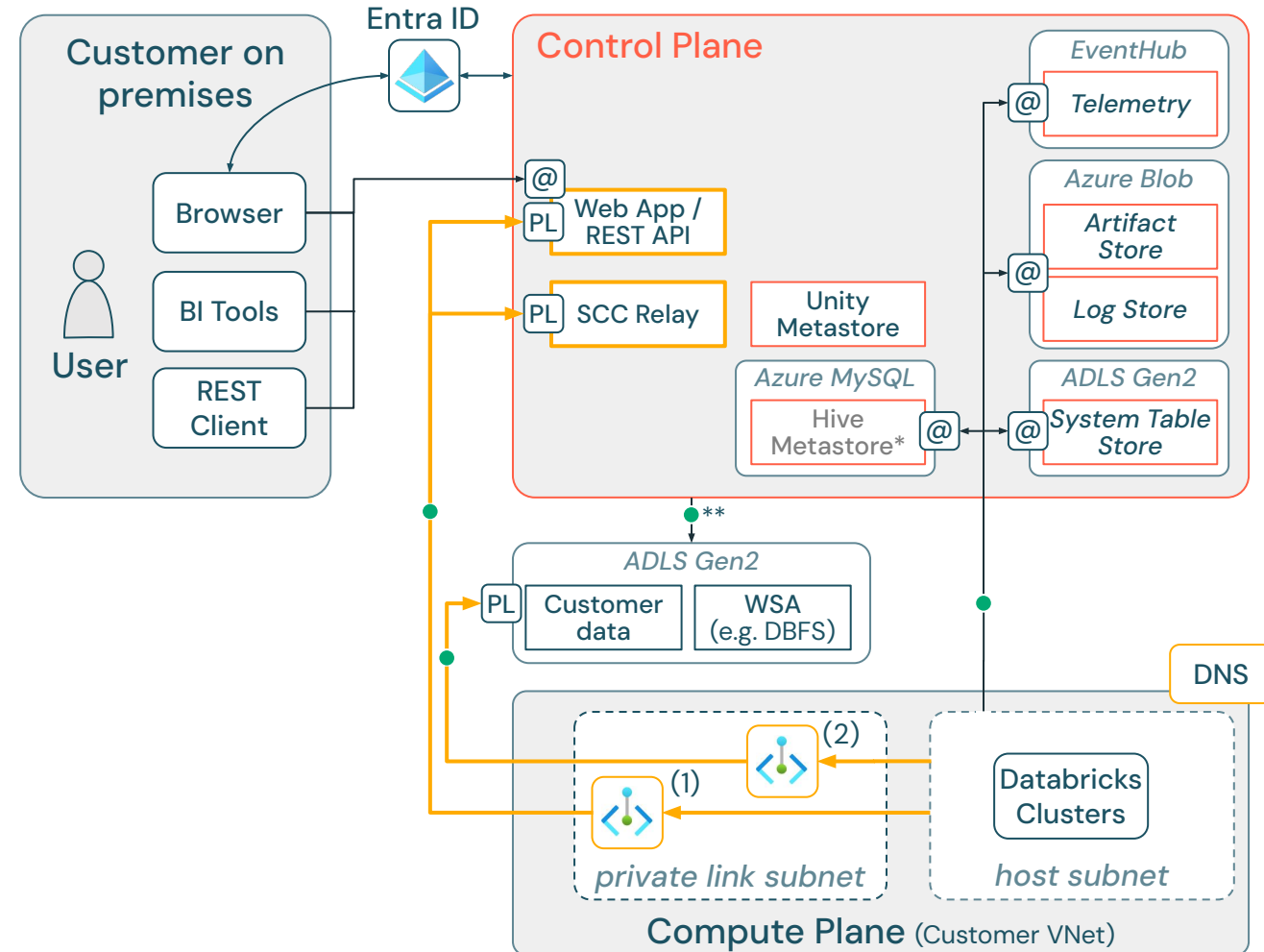
@ = Public IP (TLS encrypted) with IP access list

Private Link – Back-end Private Link

Use Case:

Connectivity from Databricks Compute Plane (in customer-managed VNet) to Control Plane

- Create a subnet for the private endpoints (do not define any NSG rules for a subnet that contains private endpoints)
- On Azure, Log / Telemetry / system tables store and Artifact Store currently cannot be put behind a PE (technical limitations)
- DBFS and customer data should be accessed via standard Private Link setups (Azure specific, not part of Databricks Private Link). Same holds for all other cloud services
- Control Plane to DBFS uses access connector and it is private
- DNS change for the Customer VNet: Route traffic to the private endpoint for the Web App
- Can be combined with the “Front-end Private Link” setup (see next slide)



● = via Azure Backbone

@ = Public IP, TLS encrypted

PL = Private Link Service

PE = Private Endpoint

(1) PE for SCC Relay & Web App

(2) PEs for storage container

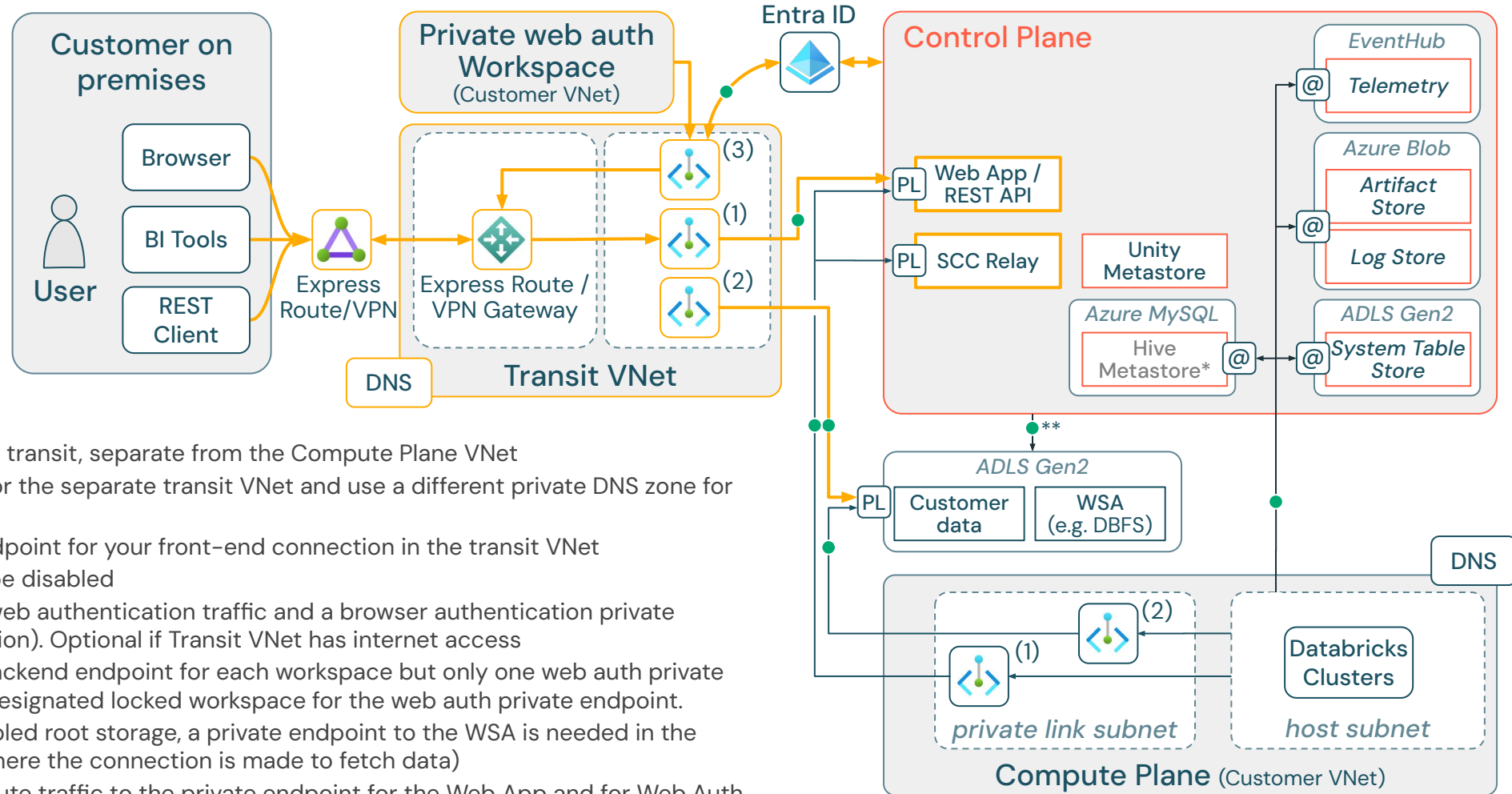
* Legacy, for migrations only

** Managed Identity, no public IP is used

Private Link – Front-end Private Link

Use Case:

Connectivity from on premises User to the Workspace endpoints (including notebooks, REST API, JDBC/ODBC, Databricks Connect)



- It is recommended to
 - create a different VNet for transit, separate from the Compute Plane VNet
 - create a resource group for the separate transit VNet and use a different private DNS zone for that private endpoint
 - use a separate private endpoint for your front-end connection in the transit VNet
- Public access via public IPs can be disabled
- For SSO, a workspace to handle web authentication traffic and a browser authentication private endpoint is required (one per region). Optional if Transit VNet has internet access
- There will be one frontend and backend endpoint for each workspace but only one web auth private endpoint for each region. Use a designated locked workspace for the web auth private endpoint.
- For Cloud Fetch with firewall enabled root storage, a private endpoint to the WSA is needed in the Transit VNet (or any Vnet from where the connection is made to fetch data)
- DNS changes for Transit VNet: Route traffic to the private endpoint for the Web App and for Web Auth

● = via Azure Backbone

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PL = Private Link Service

PE = Private Endpoint

(1) PE for SCC Relay & Web App

(2) PEs for storage container

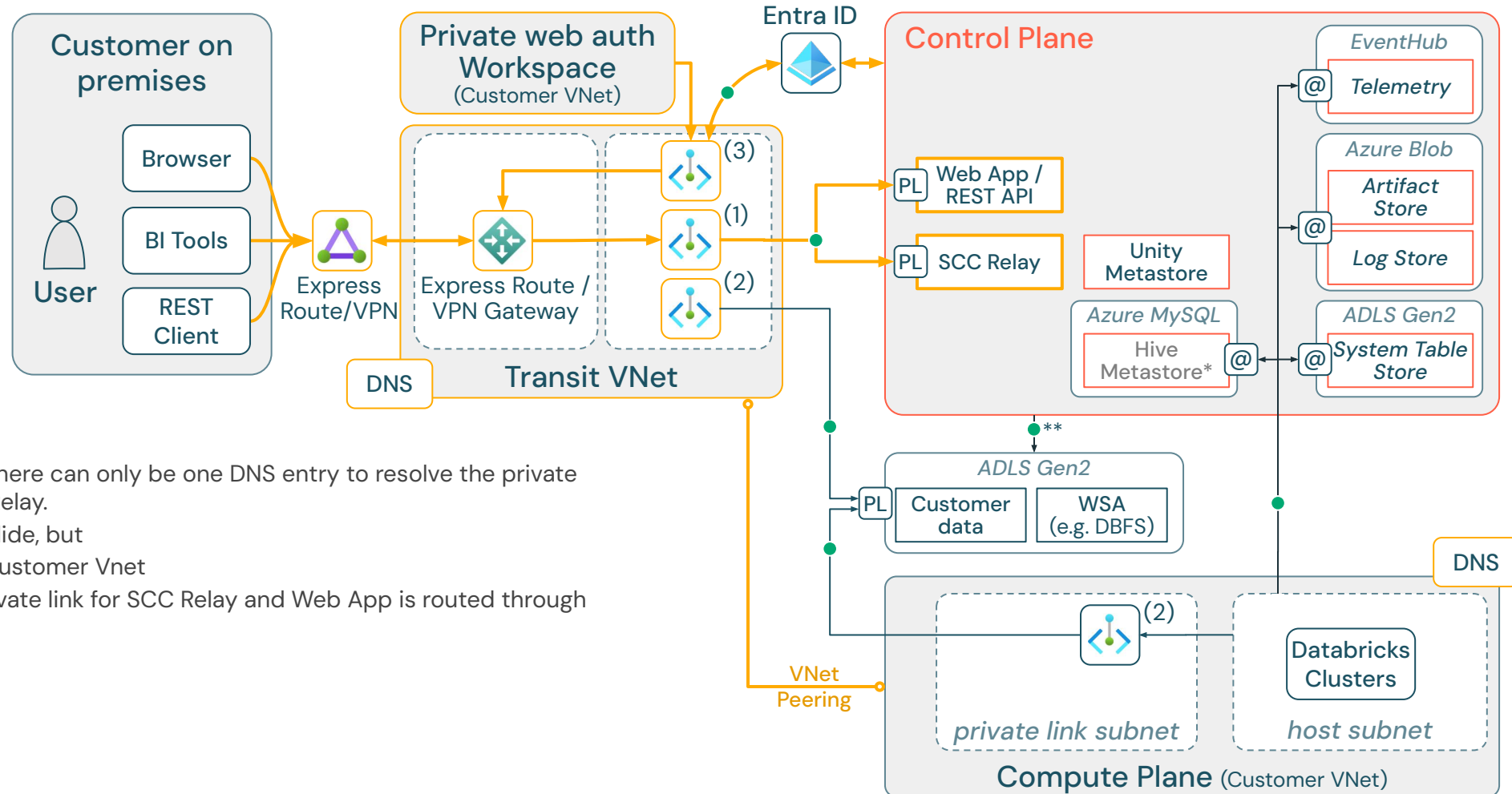
* Legacy, for migrations only

** Managed Identity, no public IP is used

Private Link – Front-end Private Link (Custom DNS)

Use Case:

Connectivity from on premises User to the Workspace endpoints with a custom DNS



- In case of a custom DNS server, there can only be one DNS entry to resolve the private endpoint for Web App and SCC Relay.
- Works like the setup on the last slide, but
 - a. Peer the Transit and the Customer Vnet
 - b. Communication to the private link for SCC Relay and Web App is routed through Transit VNet

● = via Azure Backbone
@ = Public IP, TLS encrypted
PL = Private Link Service
PE = Private Endpoint

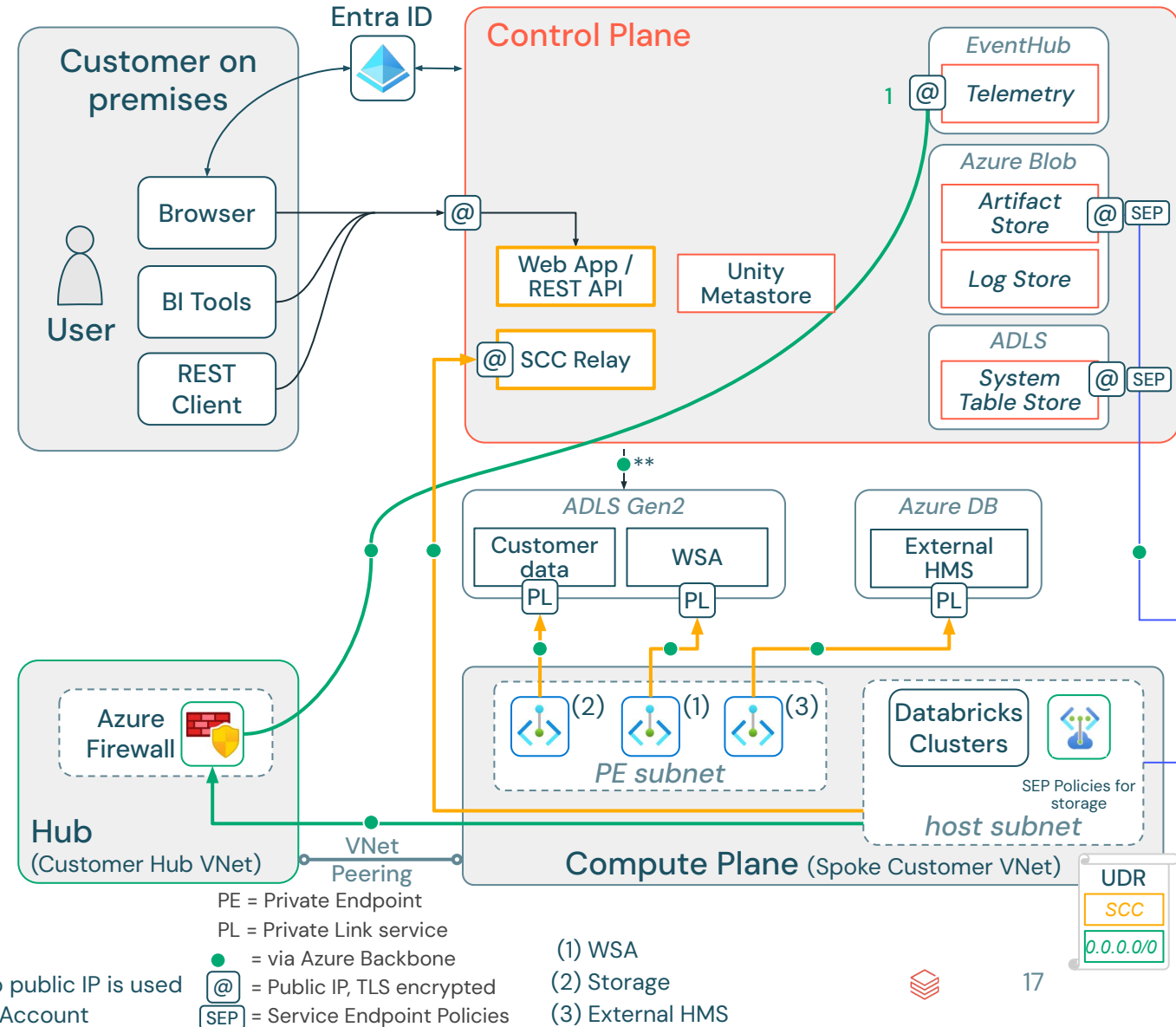
(1) PE for SCC Relay & Web App
(2) PEs for storage container
(3) PE for Web Authentication
* Legacy, for migrations only
** Managed Identity, no public IP is used

Data Exfiltration Protection w/o Databricks PL

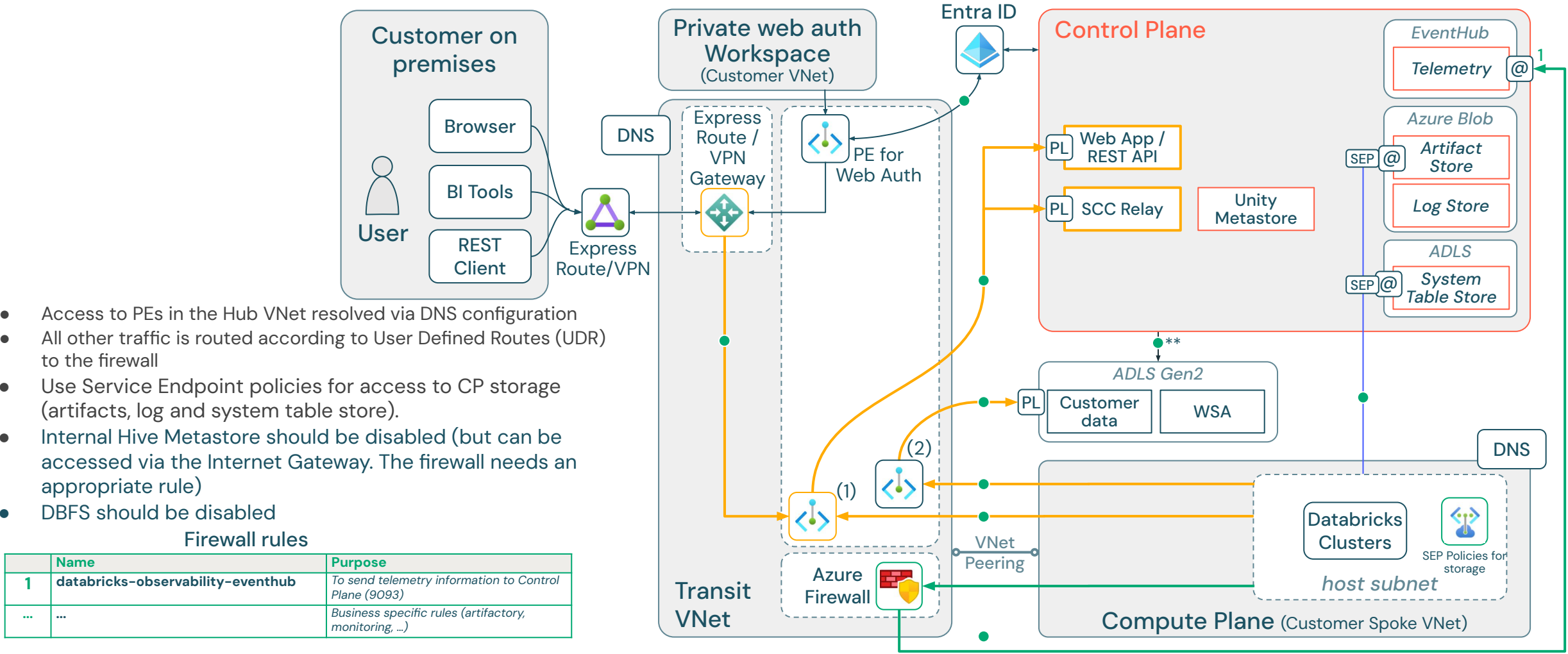
Firewall rules		
	Name	Purpose
1	databricks-observability-eventhub	To send telemetry information to Control Plane (9093)
...	...	Business specific rules (artifactory, monitoring, ...)

Notes:

- All traffic is routed according to User Defined Routes (UDR)
- It is not recommended to route the SCC connection through Azure Firewall (avoid the additional hop)
- As with the “Managed tables container”, all other storage containers should be accessed via Private Link
- Use external Metastore with private endpoint
- Use Service Endpoint policies for access to CP storage (artifacts, log and system table store)
- Control Plane to DBFS uses access connector and it's private
- This architecture can be combined with *Databricks Private Link*



Data Exfiltration Protection with Full Private Link



- Access to PEs in the Hub VNet resolved via DNS configuration
- All other traffic is routed according to User Defined Routes (UDR) to the firewall
- Use Service Endpoint policies for access to CP storage (artifacts, log and system table store).
- Internal Hive Metastore should be disabled (but can be accessed via the Internet Gateway. The firewall needs an appropriate rule)
- DBFS should be disabled

Firewall rules

	Name	Purpose
1	databricks-observability-eventhub	To send telemetry information to Control Plane (9093)
...	...	Business specific rules (artifactory, monitoring, ...)

High Level Serverless Network Architecture

Serverless Compute Plane – Networking Options

Use Case:

Connectivity from Azure Databricks Serverless Compute Plane to ADLS, external HMS (SQL DB), and other resources / internet

Databricks introduces a **Network Connectivity Configuration (NCC)** construct that assists in centrally managing network connectivity across one or multiple workspaces' Serverless Compute Planes.

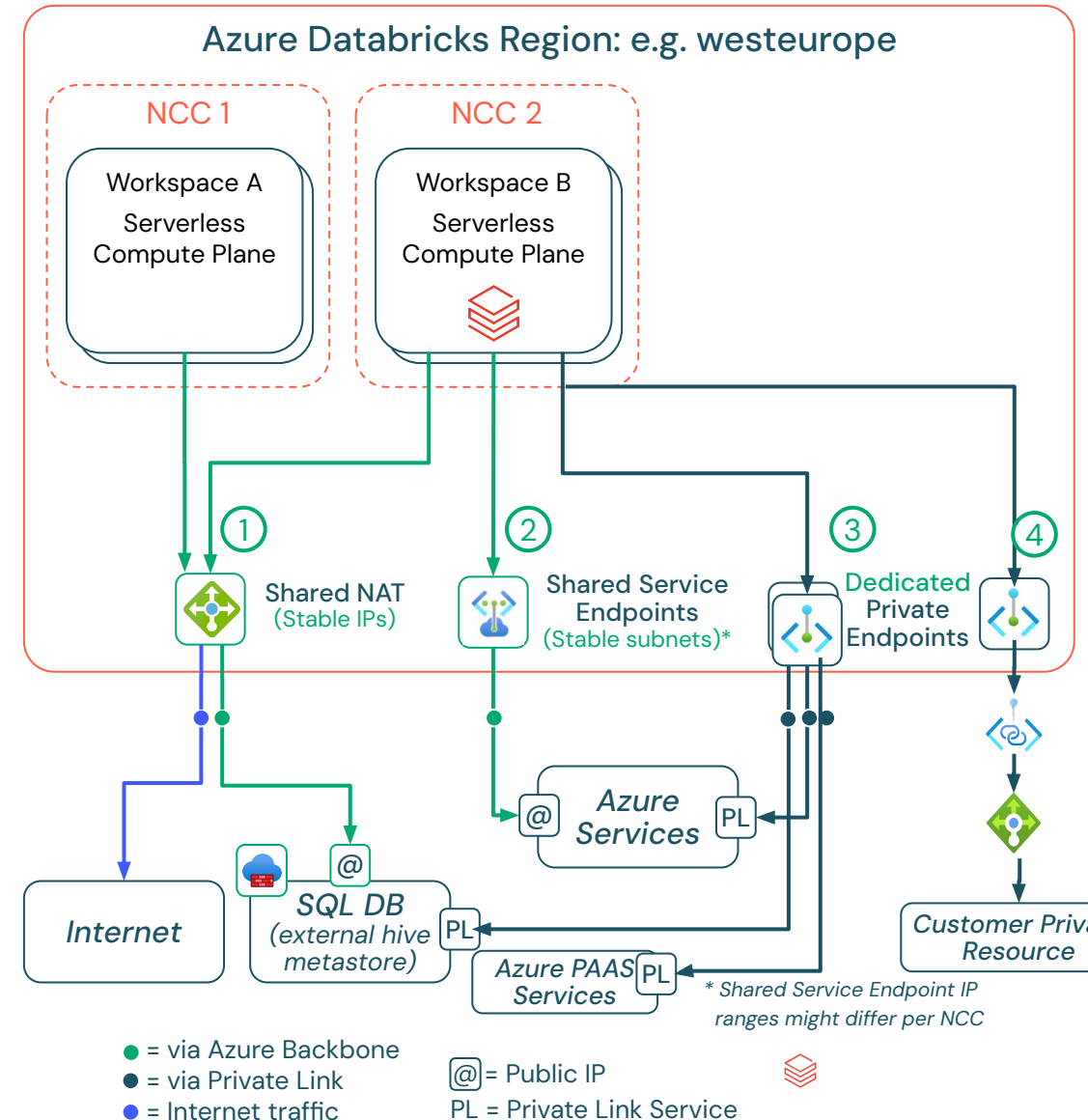
- ① With NCC, **firewall enablement** (shared NAT gateways) provides stable IPs for allowlisting on resource firewalls
- ② With NCC, **shared Service Endpoints** provide stable subnets for allowlisting on resource firewalls for e.g. Azure storage accounts, sql server and any resource that supports service endpoints
- ③ For connectivity instances where Private Link is available, you can create a dedicated Private Endpoint in the NCC for different Azure Services, which includes all resources behind a Standard Load Balancer
- ④ With NCCs, it is also possible to connect to customer resources using **dedicated private endpoint** (private link) to an endpoint service and network load balancer

Note: Availability might also depend on the cloud region

It is recommended to

- leverage direct connectivity to Azure services over an optimized route with Service Endpoints or Stable IP (where service endpoint is not supported), and only use Private Link for storage when absolutely required (to avoid additional cost)
- share an NCC among workspaces within the same business unit, environment and those sharing the same region connectivity properties.

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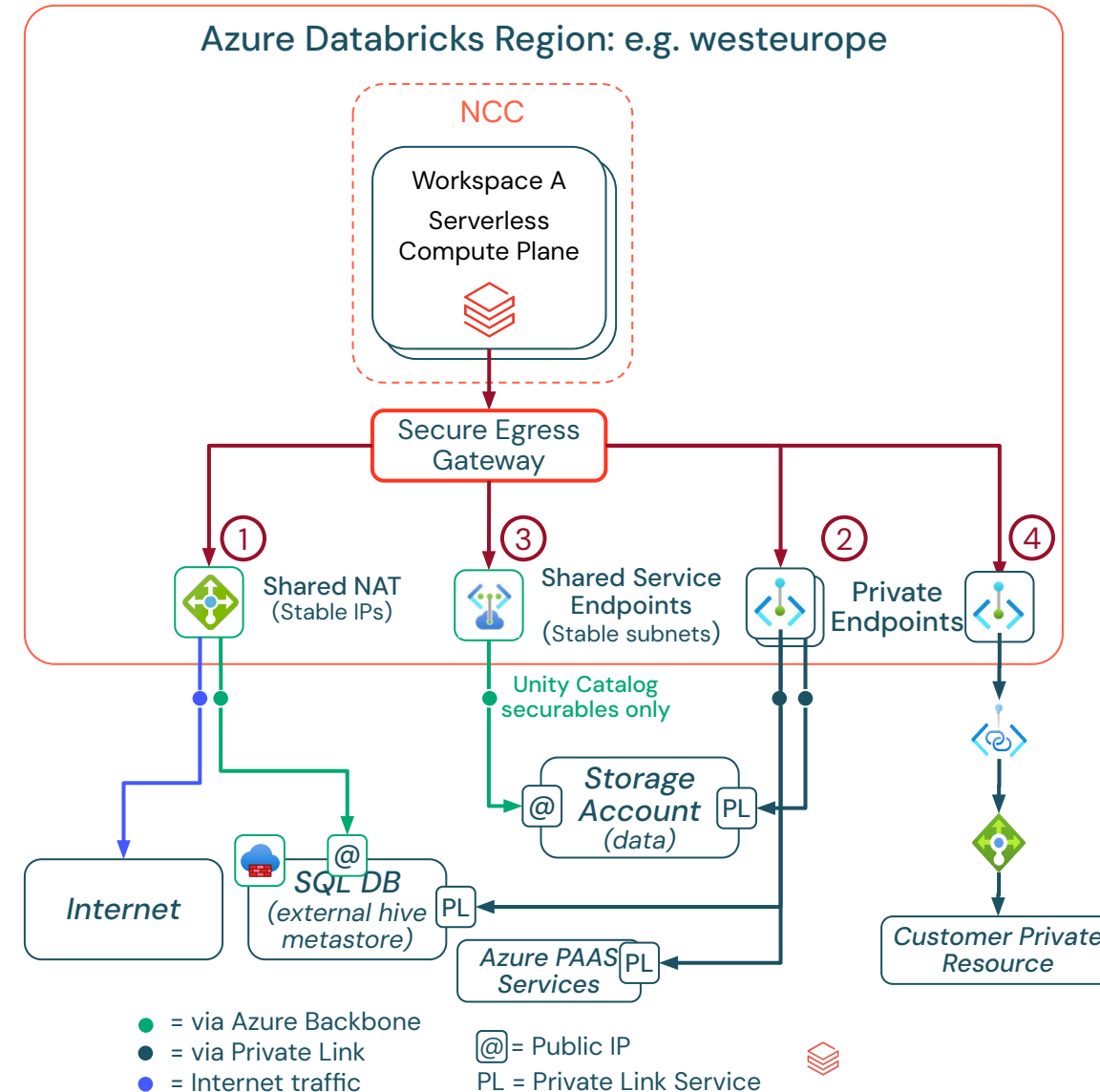
Serverless Compute Plane – Serverless Egress Control

Use Case:

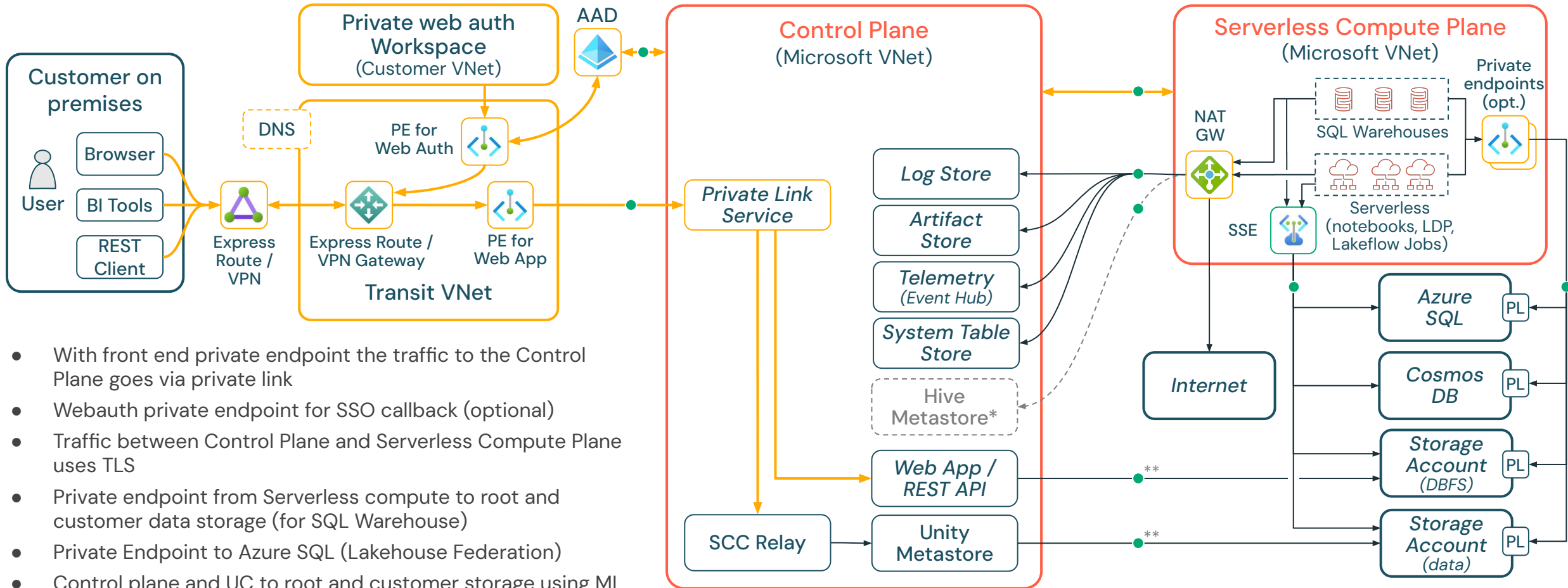
Control outbound connections from your Serverless compute resources to the internet and to cloud storage

With Serverless Egress Control you can restrict access to the internet while allowing access via Unity Catalog Connections or Private Link. Further, this feature blocks direct access to Cloud Storage to ensure that all data access occurs via Unity Catalog controlled paths.

1. Outbound access from the serverless network is denied by default. However,
 - a. access to Databricks-specific systems, e.g. the workspace root container and system tables is allowed.
 - b. FQDNs can be configured as exceptions to allow access to defined services
2. Access via private link is governed by a network policy.
3. Direct access to storage accounts from user code in UDFs and Notebooks (REPL) is denied. Access is only allowed to Unity Catalog securables



Secure Architecture with Serverless



- With front end private endpoint the traffic to the Control Plane goes via private link
- Webauth private endpoint for SSO callback (optional)
- Traffic between Control Plane and Serverless Compute Plane uses TLS
- Private endpoint from Serverless compute to root and customer data storage (for SQL Warehouse)
- Private Endpoint to Azure SQL (Lakehouse Federation)
- Control plane and UC to root and customer storage using MI
- Instead of private link to the object storage and Azure SQL, service endpoints can be used to restrict network access (for notebooks, Lakeflow Declarative Pipelines and Lakeflow Jobs)

* Legacy, for migrations only

** With Managed Identities, no public IP is used

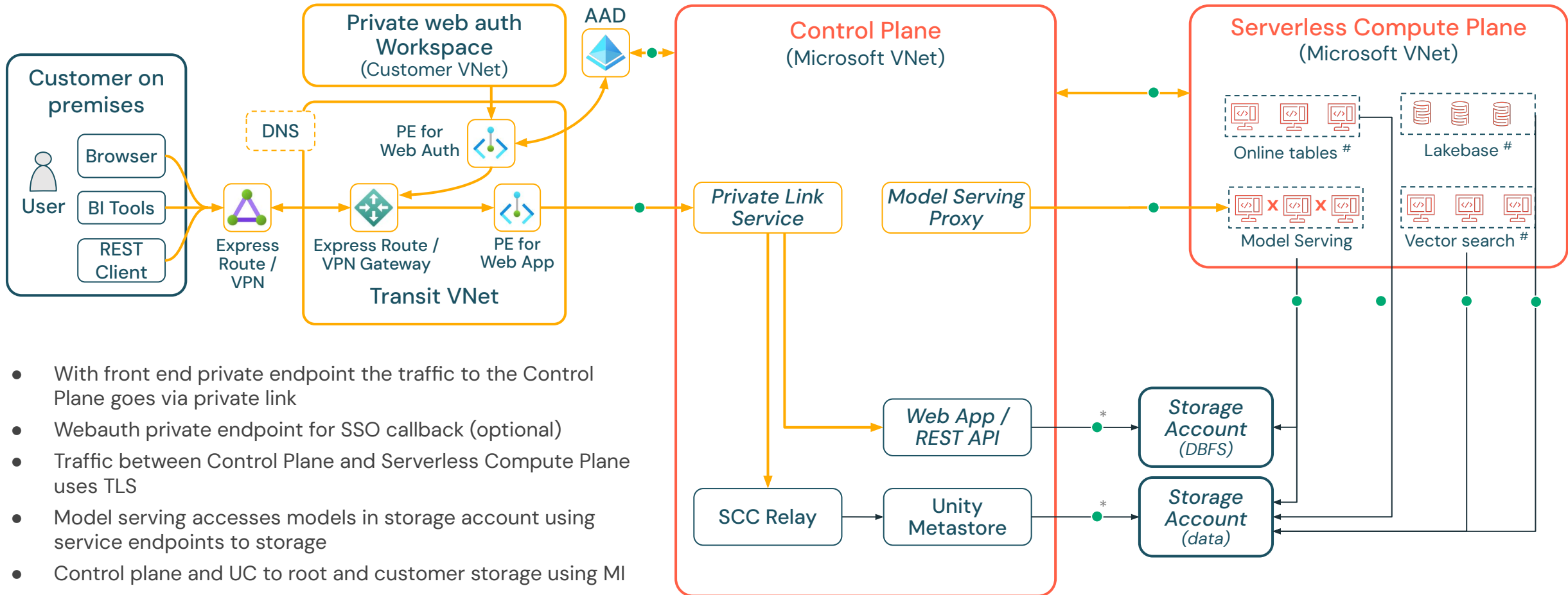
● = via Azure Backbone

@ = Public IP

SSE = Shared Service Endpoint



Secure Architecture with Serverless



- With front end private endpoint the traffic to the Control Plane goes via private link
- Webauth private endpoint for SSO callback (optional)
- Traffic between Control Plane and Serverless Compute Plane uses TLS
- Model serving accesses models in storage account using service endpoints to storage
- Control plane and UC to root and customer storage using MI
- Instead of private link to the object storage and Azure SQL, service endpoints can be used to restrict network access

* With Managed Identities, no public IP is used

Use serverless LDP in the background

x No internode traffic allowed

● = via Azure Backbone

@ = Public IP

